

Paideia Math Team Practice

Vieta's Formulas and Newton's Sums

Vieta's Formulas

For a quadratic

$$ax^2 + bx + c = 0$$

with roots s_1 and s_2 ,

$$\boxed{s_1 + s_2 = -\frac{b}{a}}, \quad \boxed{s_1 s_2 = \frac{c}{a}}.$$

For a cubic

$$a_0 x^3 + a_1 x^2 + a_2 x + a_3 = 0$$

with roots x_1, x_2, x_3 ,

$$\boxed{x_1 + x_2 + x_3 = -\frac{a_1}{a_0}}, \quad \boxed{x_1 x_2 + x_2 x_3 + x_3 x_1 = \frac{a_2}{a_0}}, \quad \boxed{x_1 x_2 x_3 = -\frac{a_3}{a_0}}.$$

The main idea is to rewrite the expression you want in terms of sums and products of roots, rather than solving for the roots individually. Two useful rewrites are

$$s_1^2 + s_2^2 = (s_1 + s_2)^2 - 2s_1 s_2, \quad \frac{1}{s_1} + \frac{1}{s_2} = \frac{s_1 + s_2}{s_1 s_2}.$$

Let m and n be the two roots of the equation

$$x^2 - 15x + 28 = 0.$$

Find $(m+1)(n+1)$.

What is the sum of the reciprocals of the roots of the equation

$$\frac{2003}{2004}x + 1 + \frac{1}{x} = 0?$$

Let a and b be the roots of the equation

$$x^2 - mx + 2 = 0.$$

Suppose that $a + \frac{1}{b}$ and $b + \frac{1}{a}$ are the roots of the equation

$$x^2 - px + q = 0.$$

What is q ?

If a, b, c , and d are nonzero numbers such that c and d are the solutions of

$$x^2 + ax + b = 0$$

and a and b are the solutions of

$$x^2 + cx + d = 0,$$

then $a + b + c + d$ equals

A. 0 **B.** -2 **C.** 2 **D.** 4 **E.** $\frac{-1 + \sqrt{5}}{2}$

Let

$$g(x) = x^3 - 5x^2 + 2x - 7,$$

and let the roots of $g(x)$ be p, q , and r . Compute

$$p^2qr + pq^2r + pqr^2.$$

Let r_1, r_2 , and r_3 be the roots of the polynomial

$$x^3 - 14x + 15x - 16.$$

Compute

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}.$$

Find the value of q if the product of two of the three roots of the equation

$$x^3 - 5x^2 + qx + 12 = 0$$

is -4 .

Newton's Sums

Consider a polynomial of degree n ,

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

with roots $x_1, x_2, \dots, x_n$. Define the power sum

$$P_k = x_1^k + x_2^k + \cdots + x_n^k.$$

Newton's sums relate these power sums directly to the coefficients of the polynomial. The first few identities are

$$a_n P_1 + a_{n-1} = 0,$$

$$a_n P_2 + a_{n-1} P_1 + 2a_{n-2} = 0,$$

$$a_n P_3 + a_{n-1} P_2 + a_{n-2} P_1 + 3a_{n-3} = 0.$$

In the same pattern,

$$\boxed{a_n P_k + a_{n-1} P_{k-1} + \cdots + a_{n-k+1} P_1 + k a_{n-k} = 0}.$$

This lets you compute sums such as $x_1^k + x_2^k + \cdots + x_n^k$ without finding the roots themselves.

If p, q , and r are distinct roots of

$$x^3 - x^2 + x - 2 = 0,$$

then $p^3 + q^3 + r^3$ equals

- A.** -1 **B.** 1 **C.** 3 **D.** 5 **E.** none of these